Unit 1: Foundations and Prerequisites

Cheat Sheet and Review Guide

Built on the Math Confidence Reset Method

First principles before formulas, clarity before memorization

Staples Education

Ordinary Differential Equations Roadmap

Contents

1 Part I: Condensed Cheat Sheet

This part distills the four core topics of Unit 1. For every topic you will find the underlying logic, the standard form to commit to working memory, and a clear step by step method. The goal is understanding you can rebuild from scratch, not symbols you simply hope to recall.

1.1 The Calculus Toolkit Review

The Core Logic

A derivative measures an instantaneous rate of change, that is, the slope of a curve at a single point. An integral measures accumulation, the running total of a quantity. Differentiation and integration are inverse operations, so each one undoes the other. Master these two ideas and every rule below becomes a shortcut rather than a mystery.

The Standard Form

Let u and v be functions of x, and let n be a constant.

$$\textbf{Power rule:}\quad \frac{d}{dx}\,x^n = n\,x^{n-1}$$

$$\textbf{Chain rule:}\quad \frac{d}{dx}\,f\big(g(x)\big) = f'\big(g(x)\big)\cdot g'(x)$$

$$\textbf{Product rule:}\quad \frac{d}{dx}\,(uv) = u'v + uv'$$

$$\textbf{Power rule for integrals:}\quad \int x^n\,dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1$$

Step-by-Step Methodology

1. Identify the structure of the expression. Ask whether it is a single power, a function nested inside another function, or a product of two functions.
2. A nested structure signals the chain rule. A multiplication of two non constant factors signals the product rule. A plain power signals the power rule.
3. Apply the matching rule one layer at a time, and never differentiate two layers in a single step.
4. When integrating, raise the exponent by one, divide by the new exponent, and then always add the constant of integration C.

Differentiating a Nested Power

Differentiate $h(x) = (x^2+1)^3$.

$$\begin{aligned}
\text{Outer function:}&\quad f(u) = u^3, \quad \text{so } f'(u) = 3u^2\\
\text{Inner function:}&\quad g(x) = x^2+1, \quad \text{so } g'(x) = 2x\\
\text{Chain rule:}&\quad h'(x) = 3(x^2+1)^2\cdot 2x\\
\text{Simplify:}&\quad h'(x) = 6x\,(x^2+1)^2
\end{aligned}$$

The logic, the rate of change of the whole equals the rate of the outer shell times the rate of the inner core.

Common Pitfall

Two errors dominate this topic. First, forgetting the inner derivative in the chain rule, which leaves the answer incomplete. Second, dropping the constant of integration C. Every indefinite integral represents a whole family of curves, and omitting C silently discards every member of that family except one.

1.2 The Number e

The Core Logic

The number e is not a random decimal near 2.718. It is the one base for which the exponential function is exactly its own derivative. This single property is the reason e appears in every model of natural growth, decay, and continuous change.

The Standard Form

$$\textbf{Limit definition:} \quad e = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n$$

$$\textbf{Continuous compounding:} \quad A = P\,e^{rt}$$

$$\textbf{Self replicating rate:} \quad \frac{dy}{dt} = y \implies y = C\,e^{t}$$

Step-by-Step Methodology

1. Read the limit definition as a process. Splitting growth into more and more frequent steps drives the result toward the fixed number e.
2. For continuous compounding, identify the principal P, the rate r, and the time t, then substitute into $A = Pe^{rt}$.
3. For a rate that is proportional to the amount present, recognize the form $dy/dt = ky$ and write the solution directly as $y = Ce^{kt}$.
4. Pin down the constant C by applying the starting value, usually the amount at time zero.

Growth Proportional to Size

Solve $\frac{dy}{dt} = 3y$ with the initial value $y(0) = 5$.

$$\begin{aligned} \text{Recognize the form:} \quad & \frac{dy}{dt} = ky \text{ with } k = 3 \\ \text{Write the general solution:} \quad & y = C\,e^{3t} \\ \text{Apply } y(0) = 5: \quad & 5 = C\,e^{0} = C \\ & y = 5\,e^{3t} \end{aligned}$$

The constant C is simply the starting amount, because e^0 equals 1.

Common Pitfall

Do not treat e as interchangeable with 2.718 in symbolic work, and do not assume that compounding more often grows money without limit. The limit definition shows the value settles at e, so continuous compounding gives a finite, predictable result rather than runaway gain.

1.3 Euler's Formula and Identity

The Core Logic

Euler's formula reveals that a complex exponential is a rotation. The expression $e^{i\theta}$ is the point on the unit circle reached by turning through an angle θ, measured in radians from the positive real axis. Complex multiplication is therefore a combination of stretching and rotating.

The Standard Form

$$\begin{aligned} \textbf{Euler's formula:} \quad & e^{i\theta} = \cos\theta + i\sin\theta \\ \textbf{Euler's identity:} \quad & e^{i\pi} + 1 = 0 \\ \textbf{Polar multiplication:} \quad & r_1 e^{i\alpha} \cdot r_2 e^{i\beta} = r_1 r_2\, e^{i(\alpha+\beta)} \end{aligned}$$

Step-by-Step Methodology

1. To evaluate $e^{i\theta}$, plug the angle into $\cos\theta + i\sin\theta$ and read off the real and imaginary parts.
2. To multiply complex numbers in polar form, multiply the magnitudes and add the angles. This is the whole content of polar multiplication.
3. Interpret multiplication by i as a rotation of ninety degrees counterclockwise, since $i = e^{i\pi/2}$.
4. Convert back to rectangular form, that is $a + bi$, only at the final step when a coordinate style answer is required.

Rotation by Multiplication

Multiply $z = 2\,e^{i\pi/6}$ by $w = 3\,e^{i\pi/3}$ and express the result in rectangular form.

$$\begin{aligned} \text{Multiply magnitudes:} \quad & 2 \cdot 3 = 6 \\ \text{Add the angles:} \quad & \frac{\pi}{6} + \frac{\pi}{3} = \frac{\pi}{6} + \frac{2\pi}{6} = \frac{\pi}{2} \\ \text{Combined polar form:} \quad & z\,w = 6\,e^{i\pi/2} \\ \text{Apply Euler's formula:} \quad & 6\left(\cos\tfrac{\pi}{2} + i\sin\tfrac{\pi}{2}\right) = 6(0 + i) \\ \text{Rectangular form:} \quad & z\,w = 6i \end{aligned}$$

Common Pitfall

Always work in radians, not degrees, when using Euler's formula, because the calculus of $e^{i\theta}$ assumes radian measure. A second frequent slip is adding magnitudes or multiplying angles. The rule is the reverse, multiply the magnitudes and add the angles.

1.4 ODE Vocabulary

The Core Logic

An ordinary differential equation is a relationship between an unknown function and its derivatives. A solution is any function that makes the relationship true everywhere it is defined. The vocabulary below lets you classify an equation at a glance, which tells you in advance which solution tools will apply.

The Standard Form

For an unknown function $y(x)$:

- **Order:** the highest derivative that appears.
- **Degree:** the power of that highest derivative once the equation is written as a polynomial in its derivatives.
- **Linear form:** $a_n(x)\,y^{(n)} + \cdots + a_1(x)\,y' + a_0(x)\,y = g(x)$, where y and all of its derivatives appear only to the first power and are never multiplied together.
- **General solution:** carries a number of arbitrary constants equal to the order.
- **Particular solution:** a general solution with the constants fixed by given conditions.

Step-by-Step Methodology

1. Scan for the highest derivative to read off the order.
2. Check the power on that highest derivative for the degree, but only after clearing roots and fractions so the equation is polynomial in its derivatives.
3. Test linearity by asking whether y and its derivatives appear only to the first power, with no products among them and no nonlinear functions wrapped around them.
4. To form an ODE by eliminating constants, differentiate the given family as many times as there are arbitrary constants, then combine the equations to remove every constant.

Forming an ODE by Eliminating a Constant

Eliminate the arbitrary constant C from the family $y = Cx^2$.

$$\begin{aligned}
\text{Start:} \quad & y = Cx^2 \\
\text{Differentiate once:} \quad & y' = 2Cx \\
\text{Solve the original for } C: \quad & C = \frac{y}{x^2} \\
\text{Substitute into } y': \quad & y' = 2\left(\frac{y}{x^2}\right)x = \frac{2y}{x} \\
\text{Resulting ODE:} \quad & x\,y' = 2y
\end{aligned}$$

One constant required one differentiation, which produced a first order equation.

Common Pitfall

Linearity refers to how y and its derivatives enter the equation, not to how the independent variable x enters. A term such as x^3y' is perfectly linear, while a term such as $y\,y'$ or $(y')^2$ is not. Also remember that degree is defined only after the equation is polynomial in its derivatives, so an equation containing $\sin(y')$ has no degree at all.

2 Part II: Review Guide

Work through all ten questions before checking any solutions. Attempt each one from first principles, and write out your reasoning rather than only the final symbol. The full worked solutions appear in Part III.

1. Differentiate $f(x) = 7x^4 - 3x^2 + 5$.

2. Differentiate $g(x) = (2x^3 + 1)^5$ using the chain rule.

3. Differentiate $h(x) = x^3 e^x$ using the product rule.

4. Evaluate $\int \left(6x^2 - 4x + 1\right)\, dx$, including the constant of integration.

5. State the limit definition of the number e, write the formula for continuous compounding, and explain in one sentence why $dy/dt = y$ is solved by an exponential function.

6. Solve the initial value problem $\dfrac{dy}{dt} = ky$ with $y(0) = y_0$.

7. State Euler's formula, then evaluate $e^{i\pi}$ and $e^{i\pi/2}$.

8. Multiply $2\,e^{i\pi/6}$ by $3\,e^{i\pi/3}$ and express the answer in rectangular form.

9. For the equation $y'' + 3y' + 2y = e^x$, state the order, the degree, and whether it is linear. Then classify $(y')^2 + y = 0$ in the same way.

10. Form the differential equation by eliminating the arbitrary